

9.8/10

EXERCISE LIST 1

① Let $J_{\mu\nu}$ be the generators of the Lorentz group.

2/2

(a) Show that the Lie algebra of the Lorentz group is

$$i[J^{\alpha\beta}, J^{\rho\sigma}] = g^{\sigma\beta} J^{\rho\alpha} + g^{\rho\alpha} J^{\sigma\beta} + g^{\rho\beta} J^{\alpha\sigma} + g^{\alpha\sigma} J^{\beta\rho}.$$

(b) Obtain the irreducible representations of the Lorentz group.

ⓐ The generators of the Lorentz group, $J^{\mu\nu}$, are such that Lorentz transformations can be expressed as $\Lambda = \exp\left(\frac{i}{2} \omega_{\mu\nu} J^{\mu\nu}\right)$ or, with matrix indexes, $\Lambda^\sigma_\rho = \left[\exp\left(\frac{i}{2} \omega_{\mu\nu} J^{\mu\nu}\right)\right]^\sigma_\rho$.

We saw in class that an infinitesimal Lorentz transformation can be written as

$$\Lambda^\mu_\nu = \delta^\mu_\nu + \omega^\mu_\nu, \text{ and } g = \Lambda^\tau g \Lambda \text{ implies that } \omega_{\mu\nu} = -\omega_{\nu\mu}.$$

An infinitesimal Lorentz transformation operator can be written as

$$U(1+\omega) \approx 1 + \frac{i}{2} \omega_{\rho\sigma} J^{\rho\sigma}, \text{ where } \omega_{\rho\sigma} = -\omega_{\sigma\rho} \text{ and } \underbrace{J^{\rho\sigma} = -J^{\sigma\rho}}_{\text{generator of the Lorentz group}}$$

Since U satisfy a composition rule,

$$U(\bar{\Lambda}) U(\Lambda) = U(\bar{\Lambda} \Lambda)$$

generator of the Lorentz group

Now we can calculate the product $U(\Lambda) U(1+\omega) U^{-1}(\Lambda)$:

$$U(\Lambda) U(1+\omega) \underbrace{U^{-1}(\Lambda)}_{U(\Lambda^{-1})} = U(\Lambda) U(1+\omega) U(\Lambda^{-1}) = U(\Lambda(1+\omega)\Lambda^{-1})$$

$$U(\Lambda) \left(1 + \frac{i}{2} \omega_{\rho\sigma} J^{\rho\sigma}\right) U^{-1}(\Lambda) = 1 + \frac{i}{2} (\Lambda \omega \Lambda^{-1})_{\mu\nu} J^{\mu\nu}$$

$$\underbrace{U(\Lambda) U(\Lambda^{-1})}_1 + U(\Lambda) \left(\frac{i}{2} \omega_{\rho\sigma} J^{\rho\sigma}\right) U^{-1}(\Lambda) = 1 + \frac{i}{2} \Lambda^\rho_\mu \omega_{\rho\sigma} \underbrace{(\Lambda^{-1})^\sigma_\nu}_{\Lambda^\sigma_\nu} J^{\mu\nu}$$

$$\text{So, } U(\Lambda) J^{\rho\sigma} U^{-1}(\Lambda) = \Lambda^\rho_\mu \Lambda^\sigma_\nu J^{\mu\nu}$$

We can now express $U(\Lambda)$ as infinitesimal transformations themselves:

$$U(\Lambda) = U(1+\omega') = 1 + \frac{i}{2} \omega'_{\alpha\beta} J^{\alpha\beta}$$

$$\left(1 + \frac{i}{2} \omega'_{\alpha\beta} J^{\alpha\beta}\right) J^{\rho\sigma} \left(1 - \frac{i}{2} \omega'_{\mu\nu} J^{\mu\nu}\right) = (\delta^\rho_\mu + \omega'^\rho_\mu) (\delta^\sigma_\nu + \omega'^\sigma_\nu) J^{\mu\nu}$$

$$\cancel{J^{\rho\sigma}} + \frac{i}{2} \underbrace{\omega'_{\alpha\beta} J^{\alpha\beta} J^{\rho\sigma}}_{(\alpha,\beta) \leftrightarrow (\mu,\nu)} - \frac{i}{2} \omega'_{\mu\nu} J^{\rho\sigma} J^{\mu\nu} + \mathcal{O}(\omega^2) = \cancel{J^{\rho\sigma}} + \omega'^\rho_\mu J^{\mu\sigma} + \omega'^\sigma_\nu J^{\rho\nu}$$

$$\frac{i}{2} [\omega'_{\mu\nu} J^{\mu\nu}, J^{\rho\sigma}] = \omega'^\rho_\mu J^{\mu\sigma} + \omega'^\sigma_\nu J^{\rho\nu}$$

Dropping the prime,

$$\frac{i}{2} [\omega_{\mu\nu} J^{\mu\nu}, J^{\rho\sigma}] = \omega_{\mu}^{\rho} J^{\mu\sigma} + \omega_{\nu}^{\sigma} J^{\rho\nu} = \omega_{\mu\nu} g^{\nu\rho} J^{\mu\sigma} - \omega_{\mu\nu} g^{\mu\sigma} J^{\rho\nu}$$

Since $\omega_{\mu\nu} = -\omega_{\nu\mu}$, given a tensor $A_{\mu\nu}$ we have that

$$A^{\mu\nu} \omega_{\mu\nu} = \frac{1}{2} (A^{\mu\nu} \omega_{\mu\nu} - \underbrace{A^{\mu\nu} \omega_{\nu\mu}}_{\mu \leftrightarrow \nu}) = \frac{1}{2} (A^{\mu\nu} \omega_{\mu\nu} - A^{\nu\mu} \omega_{\mu\nu}) = \frac{1}{2} (A^{\mu\nu} - A^{\nu\mu}) \omega_{\mu\nu}$$

Therefore

$$\frac{i}{2} [\omega_{\mu\nu} J^{\mu\nu}, J^{\rho\sigma}] = \frac{1}{2} [g^{\nu\rho} J^{\mu\sigma} - g^{\mu\rho} J^{\nu\sigma} + g^{\nu\sigma} J^{\rho\mu} - g^{\mu\sigma} J^{\rho\nu}] \omega_{\mu\nu}$$

$$\therefore i [J^{\mu\nu}, J^{\rho\sigma}] = g^{\nu\rho} J^{\mu\sigma} + g^{\mu\rho} J^{\sigma\nu} + g^{\nu\sigma} J^{\rho\mu} + g^{\mu\sigma} J^{\rho\nu}$$

or, in terms of the indices from the text,

$$i [J^{\alpha\beta}, J^{\rho\sigma}] = g^{\beta\rho} J^{\alpha\sigma} + g^{\alpha\rho} J^{\sigma\beta} + g^{\beta\sigma} J^{\rho\alpha} + g^{\alpha\sigma} J^{\rho\beta}$$

great!

(b) The generator of the Lorentz group, $J^{\mu\nu}$, contains both rotations and boosts: ^{up to} core full, "This" is not a representation, now you
 Rotations: $J^i = \frac{1}{2} \epsilon^{ijk} J^{jk}$ $\Rightarrow$ However, this representation is not irreducible, ^{didn't} used
 Boosts: $K^i = J^{i0}$ which can be seen by calculating their commu-
 tator: any representations

$$\begin{aligned} [J^i, J^j] &= \left[\frac{1}{2} \epsilon^{iab} J^{ab}, \frac{1}{2} \epsilon^{jmn} J^{mn} \right] = \frac{1}{4} \epsilon^{iab} \epsilon^{jmn} [J^{ab}, J^{mn}] \\ &= \frac{1}{4i} \epsilon^{iab} \epsilon^{jmn} (g^{nb} J^{ma} + g^{am} J^{nb} + g^{bm} J^{an} + g^{an} J^{bm}) \Rightarrow \text{since all indices are spatial, } g \rightarrow \delta \\ &= \frac{1}{4i} (\underbrace{\epsilon^{ian} \epsilon^{jmn} J^{ma}}_{\delta^{ij} \delta^{an} - \delta^{in} \delta^{aj}} + \underbrace{\epsilon^{bin} \epsilon^{jnm} J^{nb}}_{\delta^{bn} \delta^{ij} - \delta^{in} \delta^{jb}} + \underbrace{\epsilon^{iam} \epsilon^{njm} J^{an}}_{\delta^{in} \delta^{aj} - \delta^{ij} \delta^{an}} + \underbrace{\epsilon^{bin} \epsilon^{jmn} J^{bm}}_{\delta^{bi} \delta^{jm} - \delta^{bm} \delta^{ij}}) \\ &= \frac{1}{4i} (\cancel{\delta^{ij} J^{aa}} - J^{ij} + \cancel{\delta^{ij} J^{bb}} - J^{ij} + J^{ji} - \cancel{\delta^{ij} J^{aa}} + J^{ji} - \cancel{\delta^{ij} J^{bb}}) \\ &= \frac{J^{ji}}{i} = i J^{ji} = i \epsilon^{ijk} J^k \\ &\quad \uparrow i \epsilon^{ijk} (\frac{1}{2} \epsilon^{kmn} J^{mn}) = \frac{i}{2} (\delta^{im} \delta^{jn} - \delta^{in} \delta^{jm}) J^{mn} = \frac{i}{2} (J^{ij} - J^{ji}) = i J^{ij} \end{aligned}$$

$$\begin{aligned} [K^i, K^j] &= [J^{i0}, J^{j0}] = -i (\cancel{g^{i0} J^{j0}} + \cancel{g^{00} J^{ji}} + \cancel{g^{j0} J^{i0}} + \cancel{g^{00} J^{i0}}) \therefore [J^i, J^j] = i \epsilon^{ijk} J^k // \\ &= i J^{ji} = -i J^{ij} = -i \epsilon^{ijk} J^k \end{aligned}$$

$$\begin{aligned} [J^i, K^j] &= \left[\frac{1}{2} \epsilon^{imn} J^{mn}, J^{j0} \right] = \frac{1}{2} \epsilon^{imn} [J^{mn}, J^{j0}] = \frac{-i}{2} \epsilon^{imn} (g^{nj} J^{m0} + g^{mj} J^{0n} + \cancel{g^{00} J^{jm}} + \cancel{g^{00} J^{jn}}) \\ &= \frac{-i}{2} (\underbrace{\epsilon^{imj} J^{m0}}_{K^m} + \underbrace{\epsilon^{ijn} J^{0n}}_{-K^n}) = \frac{-i}{2} (\epsilon^{imj} K^m - \epsilon^{ijn} K^n) = i \epsilon^{ijk} K^k \\ &\quad \uparrow m/n \rightarrow k \therefore [J^i, K^j] = i \epsilon^{ijk} K^k \end{aligned}$$

As $[J^i, K^j] \neq 0$, this representation is not irreducible. We may define $J_k^\pm \equiv \frac{J_k \pm iK_k}{2}$ and we can check if it's irreducible: no again. There isn't any representation here

$$[J_i^+, J_j^\pm] = \frac{1}{4} [J_i + iK_i, J_j \pm iK_j] = \frac{1}{4} ([J_i, J_j] \pm i[K_i, J_j] \pm i[J_i, K_j] \mp [K_i, K_j])$$

$$= \frac{1}{4} (i\epsilon_{ijk} J_k + \epsilon_{jik} K_k \mp \epsilon_{ijk} K_k \pm i\epsilon_{ijk} J_k) = \frac{i}{4} \epsilon_{ijk} (J_k + iK_k \pm iK_k \pm J_k) \begin{cases} [J_i^+, J_j^+] = i\epsilon_{ijk} J_k^+ \\ [J_i^+, J_j^-] = 0 \end{cases}$$

Lastly,

$$[J_i^-, J_j^\pm] = \frac{1}{4} ([J_i - iK_i, J_j \pm iK_j] - i[K_i, J_j] - [J_i, K_j] + [K_i, K_j]) = \frac{i}{4} \epsilon_{ijk} (J_k - iK_k - iK_k + J_k) = i\epsilon_{ijk} J_k^-$$

$$\begin{cases} [J_i^+, J_j^+] = i\epsilon_{ijk} J_k^+ \\ [J_i^-, J_j^-] = i\epsilon_{ijk} J_k^- \\ [J_i^+, J_j^-] = 0 \end{cases} \Rightarrow$$

They aren't J_i^+ and J_i^- are the irreducible representations of the Lorentz Group. The irreducible rep. are the proper embedding of J into

a vector space, which are labeled by the values of $(J^+)^2$ and $(J^-)^2$

$(0, 0)$ $J^+(J^++1)$ $J^-(J^-+1)$
 $(1/2, 0)$
 $(0, 1/2)$

②

2/2

Consider the four-vector electromagnetic potential

$$A^\mu = (\Phi, \vec{A})$$

where Φ ($\vec{A}$) is the scalar (vector) potential. In the absence of sources the electromagnetic Lagrangian density is

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.

(a) Obtain the classical equations of motion.

(b) Obtain the energy momentum tensor that is the conserved current when we consider the symmetry $x^\mu \rightarrow x'^\mu = x^\mu + a^\mu$ where a^μ is a constant.

① $A^\mu = (\Phi, \vec{A})$ $\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$ where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$

Let's obtain the classical equations of motion:

$$\frac{\partial \mathcal{L}}{\partial A^\alpha} - \partial_\beta \left(\frac{\partial \mathcal{L}}{\partial (\partial_\beta A^\alpha)} \right) = 0 \quad \frac{\partial \mathcal{L}}{\partial A^\alpha} = \frac{\partial}{\partial A^\alpha} (\text{terms with } \partial_\mu A_\nu) = 0$$

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial (\partial_\beta A^\alpha)} &= \frac{\partial}{\partial (\partial_\beta A^\alpha)} \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \right) = \frac{\partial}{\partial (\partial_\beta A^\alpha)} \left(-\frac{1}{4} (\partial_\mu A_\nu - \partial_\nu A_\mu) (\partial^\mu A^\nu - \partial^\nu A^\mu) \right) \\ &= -\frac{1}{4} \left[(g_{\nu\beta} \delta_\alpha^\nu \delta_\mu^\beta - g_{\mu\beta} \delta_\alpha^\mu \delta_\nu^\beta) (\partial^\mu A^\nu - \partial^\nu A^\mu) + (\partial_\mu A_\nu - \partial_\nu A_\mu) (g^{\mu\beta} \delta_\alpha^\nu \delta_\beta^\mu - g^{\nu\beta} \delta_\alpha^\mu \delta_\beta^\nu) \right] \\ &= -\frac{1}{4} \left[(g_{\nu\alpha} \delta_\mu^\beta - g_{\mu\alpha} \delta_\nu^\beta) (\partial^\mu A^\nu - \partial^\nu A^\mu) + (\partial_\mu A_\nu - \partial_\nu A_\mu) (g^{\mu\beta} \delta_\alpha^\nu - g^{\nu\beta} \delta_\alpha^\mu) \right] \\ &= -\frac{1}{4} \left[\partial^\beta A_\alpha - \partial_\alpha A^\beta - \partial_\alpha A^\beta + \partial^\beta A_\alpha + \partial^\beta A_\alpha - \partial_\alpha A^\beta - \partial_\alpha A^\beta + \partial^\beta A_\alpha \right] \\ &= \partial_\alpha A^\beta - \partial^\beta A_\alpha = F_{\alpha\beta} \end{aligned}$$

$\therefore \boxed{\partial_\beta F^{\alpha\beta} = 0}$ ← equations of motion ok

obtain

② We can ^{obtain} the conserved current using Noether's theorem (proof at the end of the list).

Let's consider the symmetry $x^\mu \rightarrow x'^\mu = x^\mu + a^\mu$. The conserved current associated with this symmetry is

$$J^\mu_\nu = \underbrace{\frac{\partial \mathcal{L}}{\partial (\partial_\mu A^\rho(x))}}_{F^\mu_\rho} \frac{dA'^\rho(x)}{da^\nu} \bigg|_{a^\nu=0} + \mathcal{L} \frac{dx'^\mu}{da^\nu} \bigg|_{a^\nu=0} \delta^\mu_\nu = F^\mu_\rho \frac{dA'^\rho(x)}{da^\nu} \bigg|_{a^\nu=0} + \mathcal{L} \delta^\mu_\nu$$

F^μ_ρ , as calculated before

$$A'^\rho(x) = A^\rho(x^\alpha - a^\alpha) \Rightarrow \frac{dA'^\rho(x)}{da^\nu} \bigg|_{a^\nu=0} = \frac{d}{da^\nu} A^\rho(x^\alpha - a^\alpha) \bigg|_{a^\nu=0}$$

$$\frac{dA'^\rho(x)}{da^\nu} \bigg|_{a^\nu=0} = \frac{dA^\rho(x)}{dx^\alpha} \cdot \frac{d}{da^\nu} (x^\alpha - a^\alpha) \bigg|_{a^\nu=0} = \frac{\partial A^\rho(x)}{\partial x^\alpha} (-\delta^\alpha_\nu) = -\partial_\nu A^\rho(x)$$

$$\text{Thus, } J^\mu_\nu = -F^\mu_\rho \partial_\nu A^\rho + \mathcal{L} \delta^\mu_\nu \Rightarrow J^{\mu\nu} = g^{\alpha\nu} J^\mu_\alpha = -g^{\alpha\nu} F^\mu_\rho \partial_\alpha A^\rho + g^{\alpha\nu} \mathcal{L} \delta^\mu_\alpha$$

$$J^{\mu\nu} = -F^{\rho\mu} \partial^\nu A_\rho + \mathcal{L} g^{\mu\nu} \Rightarrow \boxed{T^{\mu\nu} = F^{\mu\alpha} \partial^\nu A_\alpha - \frac{1}{4} g^{\mu\nu} F^{\alpha\beta} F_{\alpha\beta}}$$

good $J \rightarrow T$ → the overall sign

the conserved current is the energy-momentum tensor for $g = +---$ is opposite

(Obs: the usual form of the electromagnetic stress-energy tensor is $\tilde{T}^{\mu\nu} = F^{\mu\alpha} F^\nu_\alpha - \frac{1}{4} g^{\mu\nu} F^{\alpha\beta} F_{\alpha\beta}$, which is equivalent to $T^{\mu\nu}$.)

③ Consider a real scalar field ϕ whose Lagrangian density is

2/2

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{\lambda}{4!} \phi^4.$$

A scale transformation is defined as

$$x \rightarrow x' = e^\epsilon x$$

$$\phi(x) \rightarrow \phi'(x') = e^{-d_\phi \epsilon} \phi(x)$$

where $d_\phi = 1$ is the dimension of the field ϕ . Is this transformation a symmetry? If so, what is the conserved current?

Scale transformation: $x \rightarrow x' = e^\epsilon x$

$$\phi(x) \rightarrow \phi'(x') = e^{-\text{dimension of the field} \cdot \epsilon} \phi(x)$$

(in our case $d_\phi = 1$)

Since $\partial'_\mu = \frac{\partial}{\partial x'^\mu} = \frac{\partial}{\partial (e^\epsilon x^\mu)} = e^{-\epsilon} \partial_\mu$, when we apply this transformation the Lagrangian becomes

$$\mathcal{L}' = \frac{1}{2} \partial'_\mu \phi'(x') \partial'^\mu \phi'(x') - \frac{\lambda}{4} \phi'^4(x') = \frac{1}{2} e^{-\epsilon} \partial_\mu e^{-\epsilon} \phi(x) e^{-\epsilon} \partial^\mu e^{-\epsilon} \phi(x) - \frac{\lambda}{4} (e^{-\epsilon})^4 \phi(x)^4$$

$$= e^{-4\epsilon} \left[\frac{1}{2} \partial_\mu \phi(x) \partial^\mu \phi(x) - \frac{\lambda}{4} \phi(x)^4 \right] = e^{-4\epsilon} \mathcal{L}$$

So,

$$\delta S = \int \mathcal{L}'(\phi'(x)) d^4x' - \int \mathcal{L}(\phi(x)) d^4x = \int [e^{-4\epsilon} \mathcal{L}(\phi(x)) e^{4\epsilon} - \mathcal{L}(\phi(x))] d^4x$$

$$= \int [\mathcal{L}(\phi(x)) - \mathcal{L}(\phi(x))] d^4x \Rightarrow \boxed{\delta S = 0} \quad \therefore \text{The transformation is a symmetry.}$$

We can then obtain the conserved current:

$$J^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi(x))} \frac{d\phi'(x)}{d\epsilon} \Big|_{\epsilon=0} + \mathcal{L} \frac{dx'^\mu}{d\epsilon} \Big|_{\epsilon=0} = \partial^\mu \phi(x) \frac{d\phi'(x)}{d\epsilon} \Big|_{\epsilon=0} + \mathcal{L} \frac{dx'^\mu}{d\epsilon} \Big|_{\epsilon=0}$$

$$\frac{dx'^\mu}{d\epsilon} \Big|_{\epsilon=0} = \frac{d}{d\epsilon} (e^\epsilon x^\mu) \Big|_{\epsilon=0} = e^\epsilon x^\mu \Big|_{\epsilon=0} = x^\mu$$

$$\begin{aligned} \frac{d\phi'(x)}{d\epsilon} \Big|_{\epsilon=0} &= \frac{d}{d\epsilon} [e^{-\epsilon} \phi(e^{-\epsilon} x)] \Big|_{\epsilon=0} = [-e^{-\epsilon} \phi(e^{-\epsilon} x) + e^{-\epsilon} \partial_\alpha \phi(x) (-e^{-\epsilon} x^\alpha)] \Big|_{\epsilon=0} \\ &= [-e^{-\epsilon} \phi(e^{-\epsilon} x) - e^{-2\epsilon} \partial_\alpha \phi(x) x^\alpha] \Big|_{\epsilon=0} = -(\phi(x) + x^\alpha \partial_\alpha \phi(x)) \end{aligned}$$

So,

$$J^\mu = -\partial^\mu \phi (\phi + x^\alpha \partial_\alpha \phi) + \left(\frac{1}{2} \partial_\alpha \phi \partial^\alpha \phi - \frac{1}{4!} \phi^4 \right) x^\mu$$

$$\boxed{J^\mu = -\phi \partial^\mu \phi - \partial^\mu \phi x^\alpha \partial_\alpha \phi + \left(\frac{1}{2} \partial_\alpha \phi \partial^\alpha \phi - \frac{1}{4!} \phi^4 \right) x^\mu}$$

OK

(4)

Consider a Lorentz transformation

$$x'^\mu = \Lambda^\mu_\nu x^\nu$$

An infinitesimal Lorentz transformation takes the form

$$x'^\mu = x^\mu + \delta\omega^\mu_\nu x^\nu$$

- Show that $\delta\omega_{\mu\nu} = -\delta\omega_{\nu\mu}$.
- For a rotation, what is the form of $\delta\omega_{\mu\nu}$?
- For an scalar field $\phi(x)$ obtain the conserved current associated to a Lorentz transformation.

2/2

(a) $x'^\mu = \Lambda^\mu_\nu x^\nu$ Lorentz transformation that satisfies $\Lambda^T g \Lambda = g$

Since its infinitesimal form is $x'^\mu = x^\mu + \delta\omega^\mu_\nu x^\nu = (\delta^\mu_\nu + \delta\omega^\mu_\nu) x^\nu$,

$$g_{\mu\nu} = \Lambda^\alpha_\mu g_{\alpha\beta} \Lambda^\beta_\nu = (\delta^\alpha_\mu + \delta\omega^\alpha_\mu) g_{\alpha\beta} (\delta^\beta_\nu + \delta\omega^\beta_\nu) = g_{\mu\nu} + g_{\mu\beta} \delta\omega^\beta_\nu$$

$$\cancel{g_{\mu\nu}} = \cancel{g_{\mu\nu}} + g_{\mu\beta} \delta\omega^\beta_\nu + g_{\alpha\nu} \delta\omega^\alpha_\mu + \mathcal{O}((\delta\omega)^2) \Rightarrow \boxed{\delta\omega_{\mu\nu} = -\delta\omega_{\nu\mu}} \quad \text{OK}$$

(b) A rotation R involves only spatial components, so we're only interested about the components $\delta\omega^{ij}$, $i, j = 1, 2, 3$.

As $\delta\omega^{\mu\nu} = -\delta\omega^{\nu\mu}$, $\delta\omega^{\mu\nu}$ has only 6 independent components and it can be expressed as (see exercise 1) $\{\delta\omega^{\mu\nu}\} = \begin{pmatrix} 0 & \omega^{01} & \omega^{02} & \omega^{03} \\ -\omega^{01} & 0 & \omega^{12} & \omega^{13} \\ -\omega^{02} & -\omega^{12} & 0 & \omega^{23} \\ -\omega^{03} & -\omega^{13} & -\omega^{23} & 0 \end{pmatrix} \Rightarrow \omega^{0i}$ involves time, so we don't want them
 ω^{ij} only mix spatial components $\hookrightarrow$ rotations!

Thus, for (infinitesimal) rotations

$$\{\delta\omega^{\mu\nu}\} = \omega^{12} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} + \omega^{13} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix} + \omega^{23} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}$$

or yet, defining $(\omega^{23}, \omega^{13}, \omega^{12}) = (\theta^1, -\theta^2, \theta^3)$, $\begin{cases} \delta\omega^{ij} = \epsilon^{ijk} \theta^k, & i, j \in \{1, 2, 3\} \\ \delta\omega^{0\mu} = \delta\omega^{\mu 0} = 0 \end{cases}$ OK

(c) $x'^{\mu} = x^{\mu} + \delta x^{\mu} = x^{\mu} + \underbrace{\delta\omega^{\mu}_{\nu} x^{\nu}}_{\delta x^{\mu}}$

As $\phi'(x') = \phi(x)$, $\phi'(x) = \phi(\Lambda^{-1}x) \underset{\text{infinitesimal}}{\approx} \phi(x - \delta x) \underset{\text{Taylor expansion}}{\approx} \phi(x) - \delta x^{\mu} \partial_{\mu} \phi(x)$, so

we have that $\delta\phi(x) \equiv \phi'(x) - \phi(x) = -\delta x^{\mu} \partial_{\mu} \phi(x) = -\delta\omega^{\mu}_{\rho} x^{\rho} \partial_{\mu} \phi(x)$

So, the conserved current is

$$J^{\mu} = \frac{\partial \mathcal{L}}{\partial(\partial_{\mu} \phi(x))} \delta\phi(x) + \mathcal{L} \delta x^{\mu} = -\frac{\partial \mathcal{L}}{\partial(\partial_{\mu} \phi(x))} \delta\omega^{\alpha}_{\rho} x^{\rho} \partial_{\alpha} \phi(x) + \mathcal{L} \delta\omega^{\mu}_{\rho} x^{\rho}$$

$$= -\delta\omega^{\alpha}_{\rho} x^{\rho} \left[\frac{\partial \mathcal{L}}{\partial(\partial_{\mu} \phi(x))} \partial_{\alpha} \phi(x) - \mathcal{L} \delta^{\mu}_{\alpha} \right] \quad \therefore \boxed{J^{\mu} = -\delta\omega_{\alpha\rho} x^{\rho} T^{\mu\alpha}}$$

we can identify it as the energy-momentum tensor, T^{μ}_{α}

no, it's mandatory to remove $\delta\omega$

Alternatively, we can notice that, as $\delta\omega_{\alpha\rho} = -\delta\omega_{\rho\alpha}$,

$$\delta\omega_{\alpha\rho} x^{\rho} T^{\mu\alpha} = \frac{1}{2} \left(\delta\omega_{\alpha\rho} x^{\rho} T^{\mu\alpha} - \underbrace{\delta\omega_{\rho\alpha} x^{\rho} T^{\mu\alpha}}_{\alpha \leftrightarrow \rho} \right) = \frac{1}{2} \delta\omega_{\alpha\rho} \underbrace{(x^{\rho} T^{\mu\alpha} - x^{\alpha} T^{\mu\rho})}_{\equiv \mu^{\rho\mu\alpha}}$$

and thus we have $J^{\mu} = \frac{1}{2} \delta\omega_{\alpha\rho} \mu^{\rho\mu\alpha}$

$\Rightarrow \partial_{\mu} \mu^{\rho\mu\alpha} = 0$ Strange choice of index, but no problem.

5

Consider the vector field A^μ associated to electromagnetic field. Knowing that 1.6/2

$$A'^\mu(x') = \Lambda^\mu_\nu A^\nu(x)$$

obtain the conserved quantities due to the Lorentz invariance of the theory. Use the Lagrangian density given in the first problem.

$$\mathcal{L} = -\frac{1}{4} F^{\alpha\beta} F_{\alpha\beta}$$

Lorentz invariance: $x'^\mu = \Lambda^\mu_\nu x^\nu$
 $S[A'] = S[A]$

As we saw in the last exercise,

$$x'^\mu = x^\mu + \delta x^\mu, \text{ where } \delta x^\mu = \delta \omega^\mu_\nu x^\nu$$

$$A'^\mu(x') = \Lambda^\mu_\nu A^\nu(x) = (\delta^\mu_\nu + \delta \omega^\mu_\nu) A^\nu(x)$$

$$\begin{aligned} A'^\mu(x) &= \Lambda^\mu_\nu A^\nu(\Lambda^{-1}x) = \Lambda^\mu_\nu A^\nu(x^\alpha - \delta \omega^\alpha_\beta x^\beta) \stackrel{\text{Taylor expansion}}{=} \Lambda^\mu_\nu [A^\nu(x) - \delta \omega^\alpha_\beta x^\beta \partial_\alpha A^\nu(x)] \\ &= (\delta^\mu_\nu + \delta \omega^\mu_\nu) [A^\nu(x) - \delta \omega^\alpha_\beta x^\beta \partial_\alpha A^\nu(x)] \\ &= A^\mu(x) - \delta \omega^\alpha_\beta x^\beta \partial_\alpha A^\mu(x) + \delta \omega^\mu_\nu A^\nu(x) + \mathcal{O}((\delta \omega)^2) \end{aligned}$$

$$\delta A^\mu(x) = A'^\mu(x) - A^\mu(x) = \delta \omega^\mu_\nu A^\nu(x) - \delta \omega^\alpha_\beta x^\beta \partial_\alpha A^\mu(x)$$

By Noether's theorem, the conserved current is

$$\begin{aligned} J^\mu &= \underbrace{\frac{\partial \mathcal{L}}{\partial (\partial_\mu A^\nu(x))}}_{F_\nu{}^\mu} \delta A^\nu(x) + \mathcal{L} \delta x^\mu = F_\nu{}^\mu [\delta \omega^\nu_\rho A^\rho(x) - \delta \omega^\alpha_\beta x^\beta \partial_\alpha A^\nu(x)] + \mathcal{L} \delta \omega^\mu_\nu x^\nu \\ &= F^\nu{}^\mu [\delta \omega_{\nu\rho} A^\rho(x) - \delta \omega_{\alpha\beta} x^\beta \partial^\alpha A_\nu(x)] - \frac{1}{4} F^{\rho\sigma} F_{\rho\sigma} \delta \omega^\mu_\nu x^\nu \\ &= \delta \omega_{\alpha\beta} [F^{\alpha\mu} A^\beta(x) - F^{\mu\alpha} x^\beta \partial^\alpha A_\nu(x) - \frac{1}{4} F^{\delta\sigma} F_{\delta\sigma} g^{\mu\alpha} x^\beta] \\ &= \delta \omega_{\alpha\beta} [F^{\alpha\mu} A^\beta + x^\beta (F^{\mu\nu} \partial^\alpha A_\nu - \frac{1}{4} F^{\delta\sigma} F_{\delta\sigma} g^{\mu\alpha})] = \delta \omega_{\alpha\beta} [F^{\alpha\mu} A^\beta + x^\beta T^{\mu\alpha}] \end{aligned}$$

$T^{\mu\alpha}$, as seen in exercise 2

as I commented before, there is an overall sign in T :

$$\therefore J^\mu = \delta \omega_{\alpha\beta} [F^{\alpha\mu} A^\beta + x^\beta T^{\mu\alpha}]$$

↳ almost there. The current does not depend upon $\delta \omega$, you have to factor it, and with this you have to anti-symmetrize:

$$J^\mu = \frac{\delta \omega_{\alpha\beta}}{2} \underbrace{[F^{\alpha\mu} A^\beta - F^{\beta\mu} A^\alpha + x^\beta T^{\mu\alpha} - x^\alpha T^{\mu\beta}]}_{M^{\mu\alpha\beta}}$$

⊛ Proof of Noether's theorem:

Consider a symmetry transformation

$$\begin{cases} x^\mu \rightarrow x'^\mu = x^\mu + \delta x^\mu \\ \phi_i(x) \rightarrow \phi'_i(x') = \phi_i(x) + \tilde{\delta} \phi_i(x) \end{cases} \Rightarrow S'[\Phi'(x')] = S[\Phi(x)]$$

where $\Phi(x) = (\phi_1(x), \dots, \phi_n(x))$

$$\delta S = S'[\Phi'(x')] - S[\Phi(x)] = 0 = \int_{\Omega'} \mathcal{L}(\Phi'(x'), \partial_\mu \Phi'(x')) d^4 x' - \int_{\Omega} \mathcal{L}(\Phi(x), \partial_\mu \Phi(x)) d^4 x$$

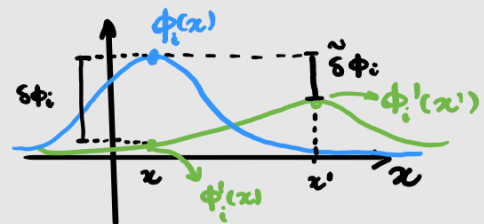
$$0 = \int_{\Omega} \left[\mathcal{L}(\Phi'(x'), \partial_\mu \Phi'(x')) \left| \det \frac{\partial x'}{\partial x} \right| - \mathcal{L}(\Phi(x), \partial_\mu \Phi(x)) \right] d^4 x$$

Now we need to work on the determinant:

$$\begin{aligned} \left| \det \frac{\partial x'}{\partial x} \right| &= \left| \det \frac{\partial}{\partial x^\nu} (x^\mu + \delta x^\mu) \right| = \left| \det (\delta^\mu_\nu + \partial_\nu \delta x^\mu) \right| \\ &= \left| \det \begin{pmatrix} 1 + \partial_0 \delta x^0 & \partial_1 \delta x^0 & \partial_2 \delta x^0 & \partial_3 \delta x^0 \\ \partial_0 \delta x^1 & 1 + \partial_1 \delta x^1 & \partial_2 \delta x^1 & \partial_3 \delta x^1 \\ \partial_0 \delta x^2 & \partial_1 \delta x^2 & 1 + \partial_2 \delta x^2 & \partial_3 \delta x^2 \\ \partial_0 \delta x^3 & \partial_1 \delta x^3 & \partial_2 \delta x^3 & 1 + \partial_3 \delta x^3 \end{pmatrix} \right| = \left| (1 + \partial_0 \delta x^0)(1 + \partial_1 \delta x^1) \right| + \mathcal{O}((\delta x)^3) \\ &= 1 + \partial_\mu \delta x^\mu + \mathcal{O}((\delta x)^2) \end{aligned}$$

Let's also define two types of variation of the field

$$\begin{cases} \text{total variation} & \tilde{\delta} \phi_i(x) = \phi'_i(x') - \phi_i(x) \\ \text{"form" variation} & \delta \phi_i(x) = \phi'_i(x) - \phi_i(x) \end{cases}$$



The two variations are related:

$$\begin{aligned} \tilde{\delta} \phi_i(x) &= \phi'_i(x') - \phi_i(x) = \phi'_i(x) + \delta x^\mu \partial_\mu \phi'_i(x) - \phi_i(x) \\ &= \delta x^\mu \partial_\mu (\delta \phi_i(x) + \phi_i(x)) + \delta \phi_i(x) = \delta x^\mu \partial_\mu \phi_i(x) + \delta \phi_i(x) + \mathcal{O}(\delta x \delta \phi) \end{aligned}$$

We also have

$$\begin{aligned} \tilde{\delta}(\partial_\mu \phi_i(x)) &= \partial'_\mu \phi'_i(x') - \partial_\mu \phi_i(x) = \frac{\partial x^\nu}{\partial x'^\mu} \frac{\partial}{\partial x^\nu} (\phi_i(x) + \tilde{\delta} \phi_i(x)) - \partial_\mu \phi_i(x) \\ &= (\delta^\nu_\mu - \partial_\mu \delta x^\nu) \partial_\nu (\phi_i(x) + \tilde{\delta} \phi_i(x)) - \partial_\mu \phi_i(x) \\ &= \cancel{\partial_\mu \phi_i(x)} + \partial_\mu \tilde{\delta} \phi_i(x) - (\partial_\mu \delta x^\nu) \partial_\nu \phi_i(x) - \cancel{\partial_\mu \phi_i(x)} + \mathcal{O}(\delta x \delta \phi) \\ &= \partial_\mu \tilde{\delta} \phi_i(x) - (\partial_\mu \delta x^\nu) \partial_\nu \phi_i(x) = \partial_\mu (\delta x^\nu \partial_\nu \phi_i(x) + \delta \phi_i(x)) - (\partial_\mu \delta x^\nu) \partial_\nu \phi_i(x) \\ &= \partial_\mu \delta \phi_i(x) + \delta x^\nu \partial_\nu \partial_\mu \phi_i(x) \end{aligned}$$

Now back to the integral,

$$0 = \int_{\Omega} \left[(1 + \partial_{\mu} \delta x^{\mu}) \underbrace{\mathcal{L}(\Phi(x'), \partial_{\mu} \Phi(x'))}_{\text{Taylor expansion with respect to } x'} - \mathcal{L}(\Phi(x), \partial_{\mu} \Phi(x)) \right] d^4 x$$

Taylor expansion with respect to x' = ①

$$\begin{aligned} \textcircled{1} &= \mathcal{L}(\Phi(x) + \tilde{\delta} \Phi(x), \partial_{\mu} \Phi(x) + \tilde{\delta}(\partial_{\mu} \Phi(x)); x + \delta x) \\ &= \mathcal{L}(\Phi(x), \partial_{\mu} \Phi(x)) + \sum_r \left[\frac{\partial \mathcal{L}}{\partial \Phi_r} \tilde{\delta} \Phi_r(x) + \frac{\partial \mathcal{L}}{\partial (\partial_{\mu} \Phi_r)} \tilde{\delta}(\partial_{\mu} \Phi(x)) \right] + \frac{\partial \mathcal{L}}{\partial x^{\mu}} \delta x^{\mu} \\ &= \mathcal{L} + \sum_r \left[\frac{\partial \mathcal{L}}{\partial \Phi_r} (\delta x^{\mu} \partial_{\mu} \Phi_r(x) + \delta \Phi_r(x)) + \frac{\partial \mathcal{L}}{\partial (\partial_{\mu} \Phi_r)} (\partial_{\mu} \delta \Phi_r(x) + \delta x^{\nu} \partial_{\nu} \partial_{\mu} \Phi_r(x)) \right] + \frac{\partial \mathcal{L}}{\partial x^{\mu}} \delta x^{\mu} \\ &= \mathcal{L} + \underbrace{(\partial_{\mu} \mathcal{L}) \delta x^{\mu}}_{\frac{d}{dx^{\mu}}} + \sum_r \frac{\partial \mathcal{L}}{\partial \Phi_r} \delta \Phi_r(x) + \underbrace{\frac{\partial \mathcal{L}}{\partial (\partial_{\mu} \Phi_r)} \partial_{\mu} \delta \Phi_r(x)}_{\partial_{\mu} \left(\frac{\partial \mathcal{L}}{\partial (\partial_{\mu} \Phi_r)} \delta \Phi_r(x) \right) - \partial_{\mu} \left(\frac{\partial \mathcal{L}}{\partial (\partial_{\mu} \Phi_r)} \right) \delta \Phi_r(x)} \\ &= \mathcal{L} + (\partial_{\mu} \mathcal{L}) \delta x^{\mu} + \partial_{\mu} \left(\sum_r \frac{\partial \mathcal{L}}{\partial (\partial_{\mu} \Phi_r)} \delta \Phi_r(x) \right) + \underbrace{\sum_r \left[\frac{\partial \mathcal{L}}{\partial \Phi_r} - \partial_{\mu} \left(\frac{\partial \mathcal{L}}{\partial (\partial_{\mu} \Phi_r)} \right) \right] \delta \Phi_r}_{=0 \text{ if } \Phi_r \text{ satisfy the equations of motion (E-L)}} \end{aligned}$$

Therefore,

$$0 = \int_{\Omega} \underbrace{\left[(\partial_{\mu} \delta x^{\mu}) \mathcal{L} + (\partial_{\mu} \mathcal{L}) \delta x^{\mu} \right]}_{\partial_{\mu} (\mathcal{L} \delta x^{\mu})} + \partial_{\mu} \left(\sum_r \frac{\partial \mathcal{L}}{\partial (\partial_{\mu} \Phi_r)} \delta \Phi_r(x) \right) + \mathcal{O}(\delta x \delta \Phi) d^4 x$$

$$\int_{\Omega} \underbrace{\partial_{\mu} \left[\mathcal{L} \delta x^{\mu} + \sum_r \frac{\partial \mathcal{L}}{\partial (\partial_{\mu} \Phi_r)} \delta \Phi_r(x) \right]}_{J^{\mu}} d^4 x = 0 \quad \forall \Omega \iff \partial_{\mu} J^{\mu} = 0$$

$$\therefore \boxed{J^{\mu} = \sum_r \frac{\partial \mathcal{L}}{\partial (\partial_{\mu} \Phi_r)} \delta \Phi_r + \mathcal{L} \delta x^{\mu}}$$

is the conserved current

great!